



BIOBANK & BIOSPECIMEN MANAGEMENT SYSTEM

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Fundamental of Databases

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Abstract

The project is building a database for donors consent details, collection events, biospecimens, aliquots, storage locations, researchers, laboratory tests and how samples are used.

The database is built using Microsoft SQL Server. The design uses primary keys and foreign keys UNIQUE constraints and CHECK constraints, indexes, views and a trigger to make sure the data is correct and to make it easier to create reports.

The SQL that was submitted also shows how to retrieve data how to join tables how to use aggregation, how to use subqueries and how to do INSERT, UPDATE and DELETE operations. It also includes testing, for data integrity.

Keywords: biobank; biospecimen; relational database; SQL Server; ER diagram; normalization; data integrity; trigger; views.

1. Problem Statement, Scope, Assumptions, and Major Business Rules

1.1 Problem Statement

Biobank must maintain accurate relationships between donor, consent, collection activities, biological specimens, storage, laboratory testing and research usage. Lack of a structured relational system can compromise data integrity through duplicate identifiers, invalid references, inconsistent domain values, and overuse of specimens. This project addresses that problem with a relational SQL Server database with explicit keys, constraints, reporting objects and automated business-rule enforcement.

1.2 Scope

The system manages donors and consents, collection events, sample and sample type management, aliquoting, storage assignment history, researcher and laboratory test requests, sample usage, reporting views, indexes, and integrity enforcement.

1.3 Assumptions

The design is based on the assumption that the consent history is maintained as multiple records, donors may participate in multiple collection events, collection events may produce multiple samples, samples may be divided into aliquots, and sample usage may occur repeatedly. The trigger assumes that quantities of usage and volumes of collection are comparable measurements.

1.4 Major Business Rules

BR-01. A donor can have multiple consent records over time.

BR-02. A donor can participate in multiple collection events.

BR-03. A collection event can produce multiple samples.

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BR-04. Each sample belongs to one sample type.

BR-05. A sample can be divided into multiple aliquots.

BR-06. An aliquot can have multiple storage assignments over time.

BR-07. A researcher can submit multiple test requests.

BR-08. Each test request identifies one sample and one test type.

BR-09. Researchers can use multiple samples, and samples can be used by multiple researchers through SAMPLE_USAGE.

BR-10. Each sample-usage record identifies one sample, one researcher, and one usage type.

BR-11. Foreign keys prevent invalid references and orphan records.

BR-12. UNIQUE constraints protect business identifiers and lookup values.

BR-13. CHECK constraints validate statuses, blood types, genders, positive quantities/volumes, and valid dates.

BR-14. The sample-usage trigger rejects cumulative usage that exceeds the sample's collection volume.

2. Project Objectives

- Keep donor and consent information.
- Record collection events and associate each event with a donor.
- Keep track of samples, sample types, aliquots, and storage history.
- Control of requests for research and laboratory tests.
- Track researchers' use of samples and the type of use.
- Ensure Referential Integrity and Domain Integrity.
- Offer reporting queries and reusable views.
- Validate the database business rules through automated testing.

3. Entities, Attributes, Identifiers, and Key Design Decisions

The system is modeled around 13 entities, each identified by a single-column primary key and described by the attributes, candidate keys, and foreign keys needed to support the business rules in Section 1.4. Table 1 in Section 4.1 lists every entity together with its attributes, primary key, foreign keys, and the key design decision behind it; the entity-relationship diagram that connects them follows immediately below.

4. Entity-Relationship Design

ER diagram illustrates the main entities and relationships of the system. It isolates donor management, specimen lifecycle management, storage, research testing, and sample use while maintaining the necessary foreign key relationships.

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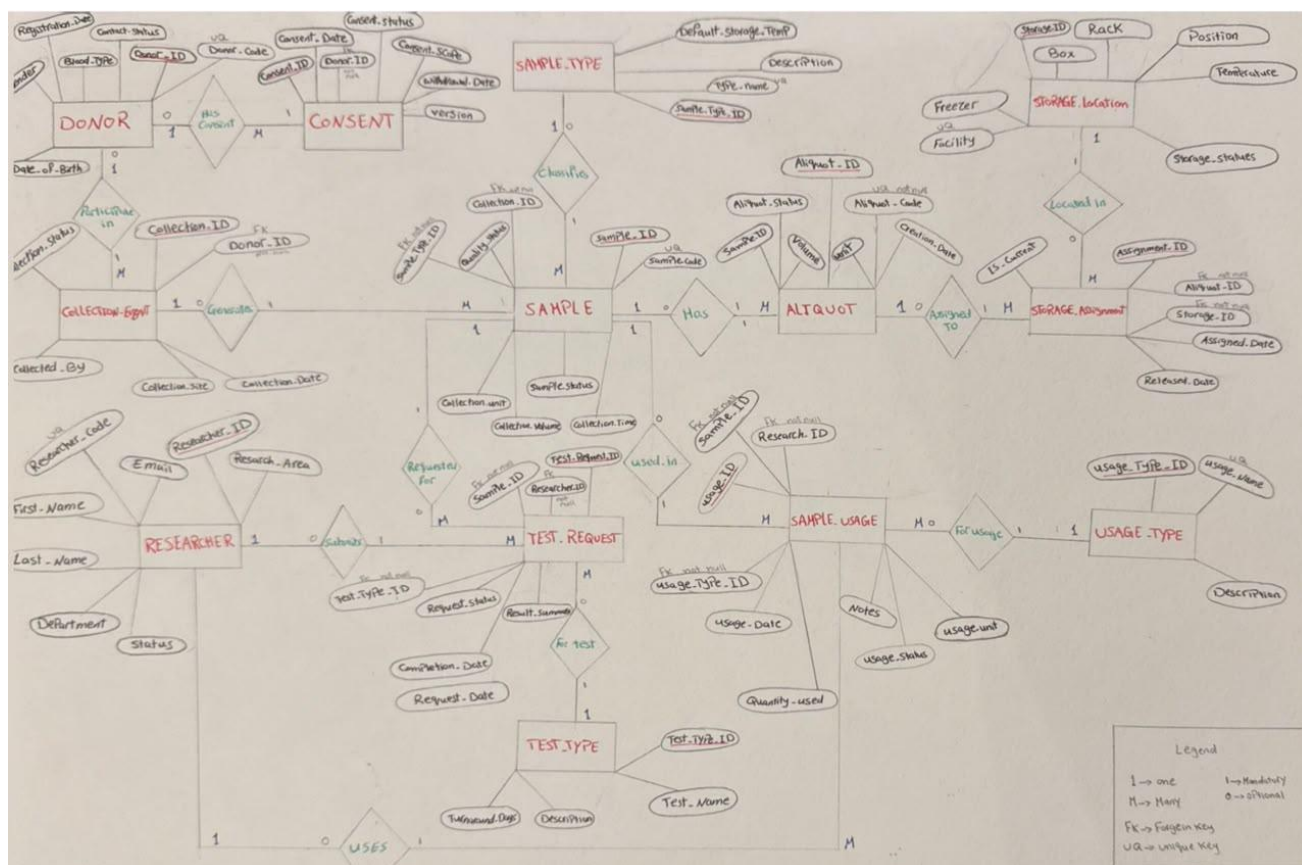


Figure 1. Entity-Relationship Diagram (ERD) for the Biobank and Biospecimen Management System.

4.1 Main Entities

The database contains 13 core entities. Table 1 lists the attributes, primary identifiers, foreign keys, and key design decisions for each relation. The design uses primary keys to uniquely identify records, foreign keys to maintain referential integrity, and UNIQUE and CHECK constraints to enforce business rules.

Entity	Attributes	Primary Key	Foreign Key(s)	Key Design Decision
DONOR	Donor_ID; Donor_Code; Date_of_Birth; Gender; Blood_Type; Registration_Date; Contact_Status	Donor_ID	—	Donor_ID is the primary identifier; Donor_Code is a unique business identifier.
CONSENT	Consent_ID; Donor_ID; Consent_Date; Consent_Scope; Consent_Version; Consent_Status; Withdrawal_Date	Consent_ID	Donor_ID → DONOR(Donor_ID)	Consent is stored separately so a donor can have multiple consent records over time.
COLLECTION_EVENT	Collection_Event_ID; Donor_ID; Collection_Date; Collection_Site;	Collection_Event_ID	Donor_ID → DONOR(Donor_ID)	Collection events are separated from donors because one

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	Collector_Name; Collection_Status			donor may have multiple events.
SAMPLE_TYPE	Sample_Type_ID; Type_Name; Description; Default_Storage_Temperature	Sample_Type_ID	—	Sample-type definitions are stored once and referenced by SAMPLE.
SAMPLE	Sample_ID; Sample_Code; Collection_Event_ID; Sample_Type_ID; Collection_Time; Collection_Volume; Status; Quality	Sample_ID	Collection_Event_ID → COLLECTION_EVENT(Collection_Event_ID); Sample_Type_ID → SAMPLE_TYPE(Sample_Type_ID)	Each sample is linked to one collection event and one sample type.
ALIQOT	Aliquot_ID; Sample_ID; Aliquot_Code; Volume; Status; Creation_Date	Aliquot_ID	Sample_ID → SAMPLE(Sample_ID)	Aliquots represent portions derived from a sample and can be managed independently.
STORAGE_LOCATION	Storage_ID; Facility; Freezer; Rack; Box; Position; Temperature; Status	Storage_ID	—	Physical storage information is separated from assignment history.
STORAGE_ASSIGNMENT	Assignment_ID; Aliquot_ID; Storage_ID; Start_Date; End_Date; Is_Current	Assignment_ID	Aliquot_ID → ALIQUOT(Aliquot_ID); Storage_ID → STORAGE_LOCATION(Storage_ID)	This associative relation preserves current and historical storage assignments.
RESEARCHER	Researcher_ID; Researcher_Code; First_Name; Last_Name; Department; Email; Research_Area; Status	Researcher_ID	—	Researcher_ID is the primary identifier; Researcher_Code and Email are protected by uniqueness rules.
TEST_TYPE	Test_Type_ID; Test_Name; Description; Turnaround_Days	Test_Type_ID	—	Test definitions are stored independently so they are not repeated in test requests.
TEST_REQUEST	Test_Request_ID; Researcher_ID; Sample_ID; Test_Type_ID; Request_Date; Completion_Date; Status	Test_Request_ID	Researcher_ID → RESEARCHER(Researcher_ID); Sample_ID → SAMPLE(Sample_ID); Test_Type_ID → TEST_TYPE(Test_Type_ID)	Each request links one researcher, one sample, and one test type.
USAGE_TYPE	Usage_Type_ID; Usage_Name; Description	Usage_Type_ID	—	Usage categories are maintained as a lookup relation.
SAMPLE_USAGE	Usage_ID; Sample_ID; Researcher_ID; Usage_Type_ID; Quantity_Used; Usage_Date; Status; Notes	Usage_ID	Sample_ID → SAMPLE(Sample_ID); Researcher_ID → RESEARCHER(Researcher_ID); Usage_Type_ID → USAGE_TYPE(Usage_Type_ID)	This associative transaction relation resolves researcher-sample many-to-many usage.

The conceptual ER model is mapped to the relational schema using the standard entity-to-relation and relationship-to-foreign-key mapping rules. Each entity is assigned a single-column primary key, and this key is propagated as a foreign key into every relation that participates in a 1:N relationship with it.

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Some relationships carry their own descriptive attributes and are therefore promoted to full associative relations rather than being represented only through a foreign key. `STORAGE_ASSIGNMENT` and `SAMPLE_USAGE` are examples: each links two entities (an aliquot to a storage location, and a researcher to a sample) while also recording attributes specific to that link, such as assignment dates and usage quantities.

`SAMPLE_TYPE`, `TEST_TYPE`, and `USAGE_TYPE` are classification (lookup) entities. Keeping them as separate relations, referenced by foreign key rather than repeated as free text, avoids storing the same descriptive values in every transaction record and keeps the schema consistent with the normalization goals discussed in Section 7.

4.2 Key Relationships

Cardinality and Participation Constraints

The ER design uses 1:N relationships throughout: `DONOR-CONSENT`, `DONOR-COLLECTION_EVENT`, `COLLECTION_EVENT-SAMPLE`, `SAMPLE_TYPE-SAMPLE`, `SAMPLE-ALIQUOT`, `ALIQUOT-STORAGE_ASSIGNMENT`, `STORAGE_LOCATION-STORAGE_ASSIGNMENT`, `RESEARCHER-TEST_REQUEST`, `SAMPLE-TEST_REQUEST`, `TEST_TYPE-TEST_REQUEST`, `RESEARCHER-SAMPLE_USAGE`, `SAMPLE-SAMPLE_USAGE`, and `USAGE_TYPE-SAMPLE_USAGE`.

In every case the child relation participates with total (mandatory) participation, since its foreign key cannot be null; the parent relation participates with partial (optional) participation, since it may have zero, one, or many related child records unless a specific business rule restricts it.

- A donor can have multiple consent records over time.
- A donor can participate in multiple collection events.
- A collection event can produce multiple samples.
- Each sample belongs to one sample type.
- A sample can be divided into multiple aliquots.
- An aliquot can have multiple storage assignments over time.
- A researcher can submit multiple test requests.
- Each test request identifies one sample and one test type.
- Researchers can use multiple samples, and samples can be used by multiple researchers through `SAMPLE_USAGE`.
- Each sample-usage record identifies one sample, one researcher, and one usage type.

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5. Relational Schema and Integrity Constraints

The database consists of 13 tables. Each table has a primary key that uniquely identifies its rows, and foreign keys that enforce referential integrity between related tables. UNIQUE constraints protect business identifiers and lookup values from duplication, while CHECK constraints validate domain rules such as status codes, blood types, gender values, positive quantities and volumes, and valid date ranges. Together, these keys and constraints form the relational schema that organizes the database and keeps its data consistent.

5.1 Relational Database Schema

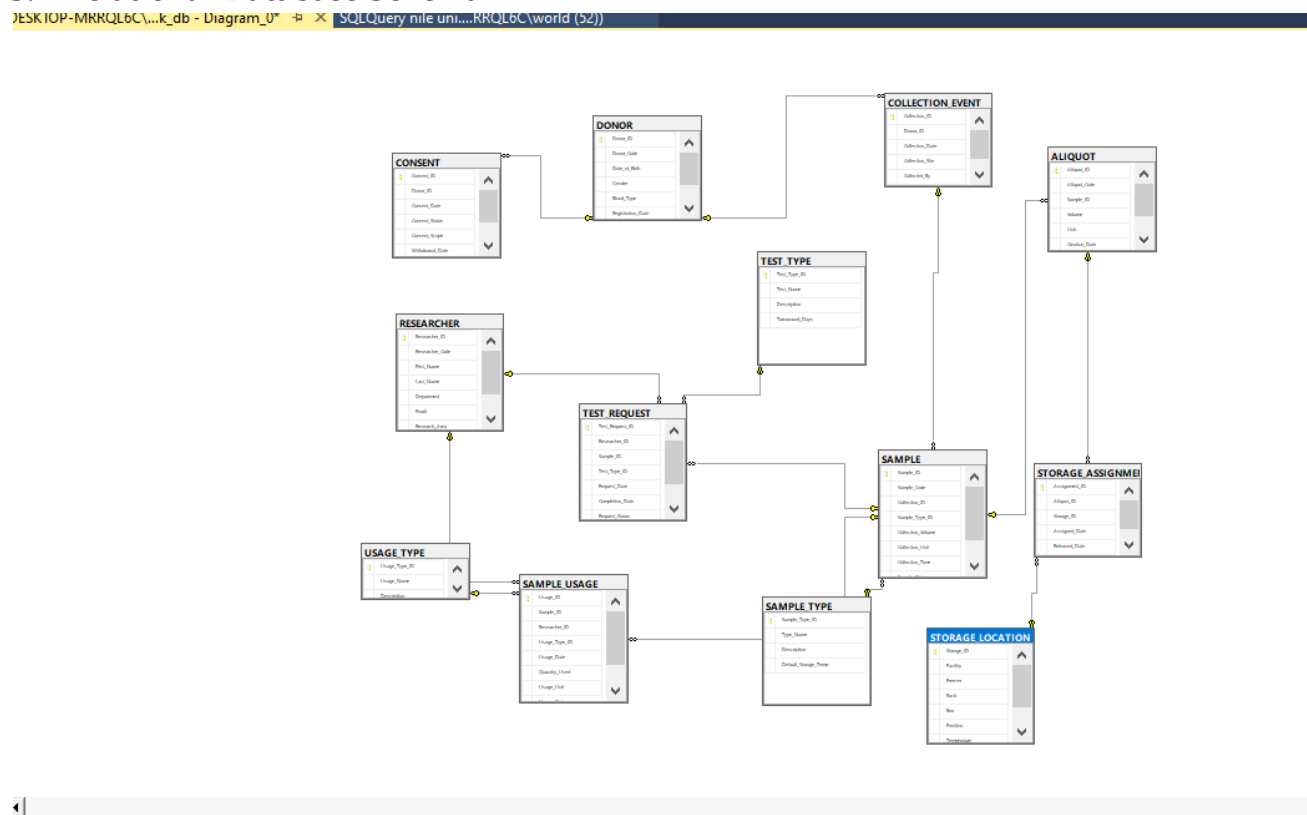


Figure 2. Relational Database Schema of the Biobank and Biospecimen Management System.

Figure 2 presents the implemented relational database schema, showing the 13 relations, their primary keys, foreign-key links, and the mapping of the conceptual design into relational tables. The textual schema summary below makes the PK/FK structure explicit; UNIQUE and CHECK constraints are described in Section 5.2.

Relation	Primary Key	Foreign Key(s)	Role / Key
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			Decision
DONOR	Donor_ID	—	Donor master relation; Donor_Code is a business identifier.
CONSENT	Consent_ID	Donor_ID → DONOR(Donor_ID)	Consent history for each donor.
COLLECTION_EVENT	Collection_Event_ID	Donor_ID → DONOR(Donor_ID)	Collection events associated with donors.
SAMPLE_TYPE	Sample_Type_ID	—	Specimen category / lookup relation.
SAMPLE	Sample_ID	Collection_Event_ID → COLLECTION_EVENT(Collection_Event_ID); Sample_Type_ID → SAMPLE_TYPE(Sample_Type_ID)	Collected biospecimen relation.
ALIQUOT	Aliquot_ID	Sample_ID → SAMPLE(Sample_ID)	Portions derived from samples.
STORAGE_LOCATION	Storage_ID	—	Physical storage- location relation.
STORAGE_ASSIGNMENT	Assignment_ID	Aliquot_ID → ALIQUOT(Aliquot_ID); Storage_ID → STORAGE_LOCATION(Storage_ID)	Historical/current storage assignments.
RESEARCHER	Researcher_ID	—	Researcher master relation; Researcher_Code is a business identifier.
TEST_TYPE	Test_Type_ID	—	Laboratory test definition / lookup relation.
TEST_REQUEST	Test_Request_ID	Researcher_ID → RESEARCHER(Researcher_ID); Sample_ID → SAMPLE(Sample_ID); Test_Type_ID → TEST_TYPE(Test_Type_ID)	Test requests linking researchers, samples, and test types.
USAGE_TYPE	Usage_Type_ID	—	Sample-usage classification / lookup relation.
SAMPLE_USAGE	Usage_ID	Sample_ID → SAMPLE(Sample_ID); Researcher_ID → RESEARCHER(Researcher_ID); Usage_Type_ID → USAGE_TYPE(Usage_Type_ID)	Associative transaction relation for researcher sample usage.

5.1.1 Database Objects

Object	Count	Details
--------	-------	---------

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Tables	13	Core biobank, storage, research, testing, and usage entities.
Indexes	3	Collection event by donor; sample by type; test request by researcher.
Views	2	vw_DonorSampleSummary and vw_TestRequestDetails.
Trigger	1	trg_CheckSampleUsageQuantity.

5.2 Integrity Constraints

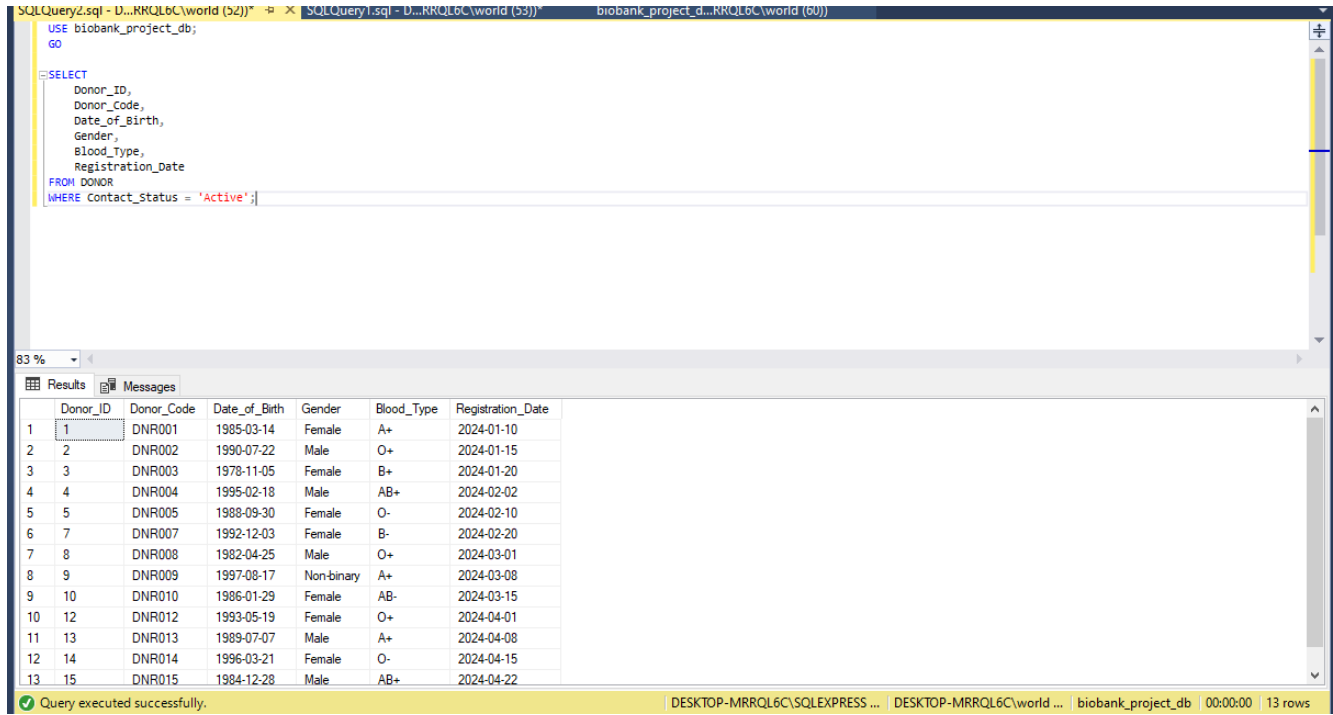
- Primary keys uniquely identify records.
- Foreign keys prevent invalid references and orphan records.
- UNIQUE constraints protect codes, email addresses, sample/aliquot codes, and lookup names.
- CHECK constraints validate statuses, blood types, genders, positive volumes/quantities, and valid dates.
- A UNIQUE storage-position constraint prevents duplicate physical positions.
- STORAGE_ASSIGNMENT.Is_Current defaults to 1.

6. Implementation Details and Selected SQL

The SQL script recreates the database, creates tables in dependency order, inserts representative data, verifies record counts, and then adds reporting and integrity objects.

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6.1 Retrieval Queries



The screenshot shows a SQL query window with the following code:

```
USE biobank_project_db;
GO

SELECT
    Donor_ID,
    Donor_Code,
    Date_of_Birth,
    Gender,
    Blood_Type,
    Registration_Date
FROM DONOR
WHERE Contact_Status = 'Active';
```

The results pane displays 13 rows of data for active donors. The status bar at the bottom indicates "Query executed successfully." and "13 rows".

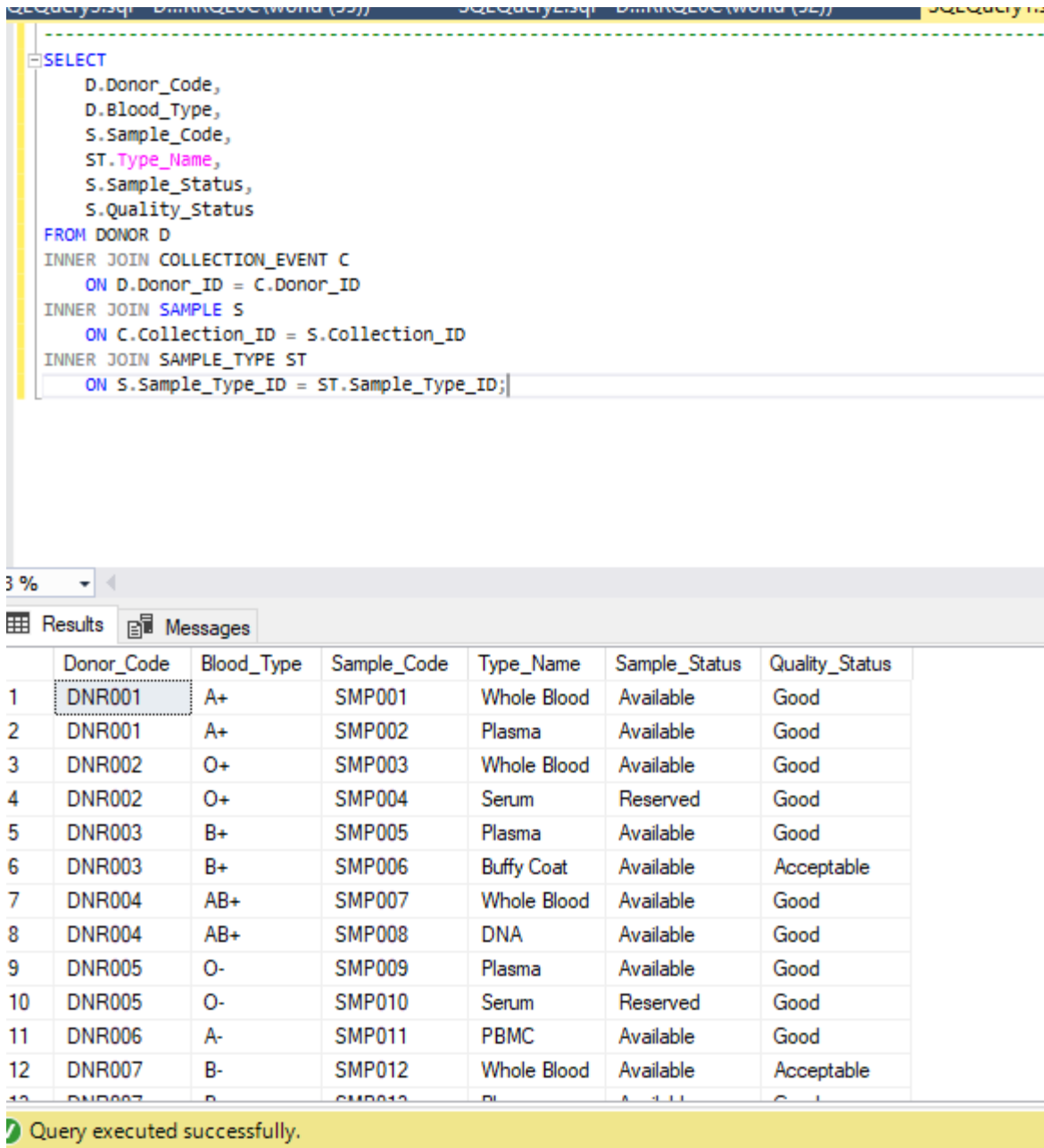
	Donor_ID	Donor_Code	Date_of_Birth	Gender	Blood_Type	Registration_Date
1	1	DNR001	1985-03-14	Female	A+	2024-01-10
2	2	DNR002	1990-07-22	Male	O+	2024-01-15
3	3	DNR003	1978-11-05	Female	B+	2024-01-20
4	4	DNR004	1995-02-18	Male	AB+	2024-02-02
5	5	DNR005	1988-09-30	Female	O-	2024-02-10
6	7	DNR007	1992-12-03	Female	B-	2024-02-20
7	8	DNR008	1982-04-25	Male	O+	2024-03-01
8	9	DNR009	1997-08-17	Non-binary	A+	2024-03-08
9	10	DNR010	1986-01-29	Female	AB-	2024-03-15
10	12	DNR012	1993-05-19	Female	O+	2024-04-01
11	13	DNR013	1989-07-07	Male	A+	2024-04-08
12	14	DNR014	1996-03-21	Female	O-	2024-04-15
13	15	DNR015	1984-12-28	Male	AB+	2024-04-22

Figure 3. SELECT Query Result Showing Active Donors.

- All donors.
- Active donors.
- Good-quality samples.
- Researchers in Molecular Biology.

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6.2 Join Queries



The screenshot displays a SQL query in a text editor and its execution results in a table. The query is a JOIN query linking Donors, Samples, and Sample Types.

```

SELECT
    D.Donor_Code,
    D.Blood_Type,
    S.Sample_Code,
    ST.Type_Name,
    S.Sample_Status,
    S.Quality_Status
FROM DONOR D
INNER JOIN COLLECTION_EVENT C
    ON D.Donor_ID = C.Donor_ID
INNER JOIN SAMPLE S
    ON C.Collection_ID = S.Collection_ID
INNER JOIN SAMPLE_TYPE ST
    ON S.Sample_Type_ID = ST.Sample_Type_ID;

```

The results table shows 12 rows of data, with the first row highlighted. The columns are Donor_Code, Blood_Type, Sample_Code, Type_Name, Sample_Status, and Quality_Status.

	Donor_Code	Blood_Type	Sample_Code	Type_Name	Sample_Status	Quality_Status
1	DNR001	A+	SMP001	Whole Blood	Available	Good
2	DNR001	A+	SMP002	Plasma	Available	Good
3	DNR002	O+	SMP003	Whole Blood	Available	Good
4	DNR002	O+	SMP004	Serum	Reserved	Good
5	DNR003	B+	SMP005	Plasma	Available	Good
6	DNR003	B+	SMP006	Buffy Coat	Available	Acceptable
7	DNR004	AB+	SMP007	Whole Blood	Available	Good
8	DNR004	AB+	SMP008	DNA	Available	Good
9	DNR005	O-	SMP009	Plasma	Available	Good
10	DNR005	O-	SMP010	Serum	Reserved	Good
11	DNR006	A-	SMP011	PBMC	Available	Good
12	DNR007	B-	SMP012	Whole Blood	Available	Acceptable

Query executed successfully.

Figure 4. JOIN Query Result Linking Donors, Samples, and Sample Types.

The submitted joins combine donors with collection events, samples with sample types, researchers with test requests, donors with samples and sample types, and samples with complete usage information.

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6.3 Aggregation

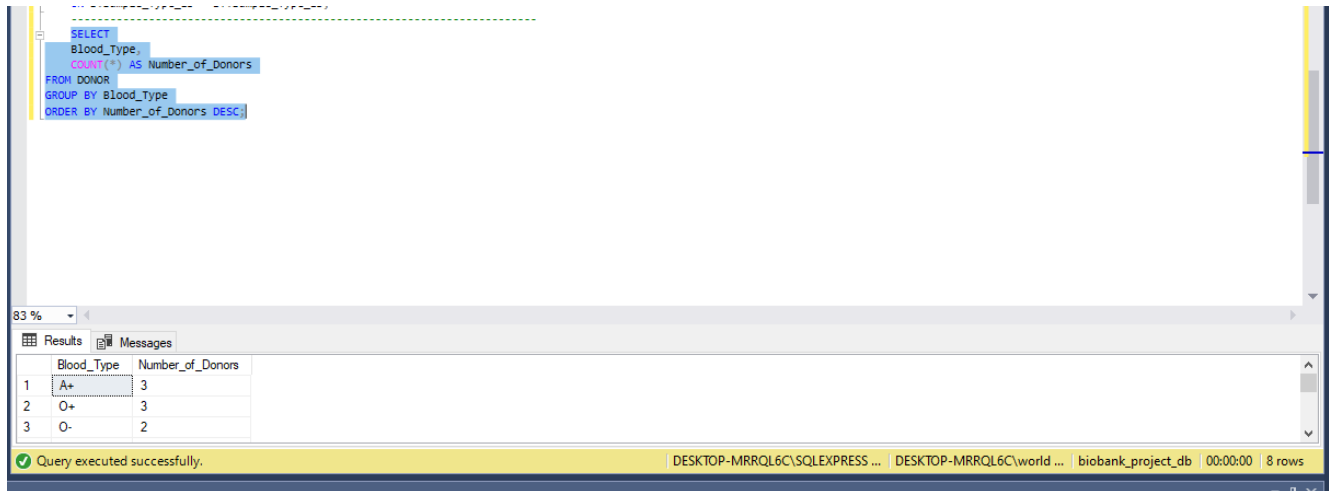


Figure 5. Aggregation Query Result Showing the Number of Donors by Blood Type.

- Count total donors.
- Count donors by blood type.
- Count samples by sample type.
- Calculate average collection volume by sample type.
- Calculate total quantity used by each researcher.
- Find maximum and minimum sample volume.
- Count test requests by status.

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6.4 Subqueries

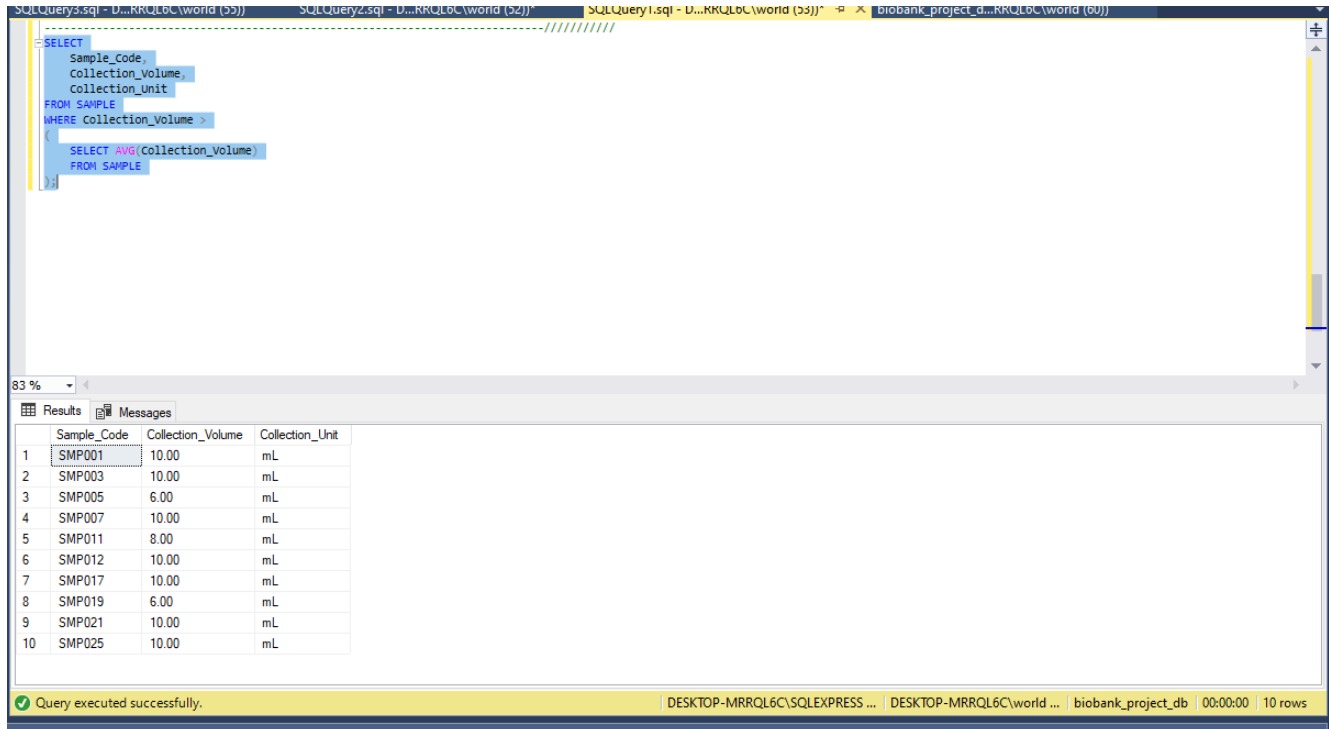


Figure 6. Subquery Result Showing Samples with Collection Volumes Above the Overall Average.

- Find donors who have at least one sample.
- Find researchers who have submitted test requests.
- Find samples above the average collection volume.
- Find researchers whose total sample usage is above the average researcher total.

7. Normalization to Third Normal Form (3NF)

First Normal Form (1NF): It makes sure that the design uses attributes and separates repeating business subjects into rows and related tables. For example consent history, sample types, test types, usage types and storage assignments are all modeled as relations rather than repeating groups.

Second Normal Form (2NF): Every table in the schema uses a single-column surrogate primary key (for example Sample_ID, Aliquot_ID, Usage_ID), so no table has a composite primary key and partial-key dependency on part of the key cannot occur. Descriptive attributes that depend only on a lookup value, rather than on the transaction record as a whole, are factored out into their own relations: sample-type descriptions live in SAMPLE_TYPE, test definitions in TEST_TYPE, usage classifications in USAGE_TYPE, and storage-location properties in STORAGE_LOCATION, instead of being repeated in SAMPLE, TEST_REQUEST, SAMPLE_USAGE, or STORAGE_ASSIGNMENT.

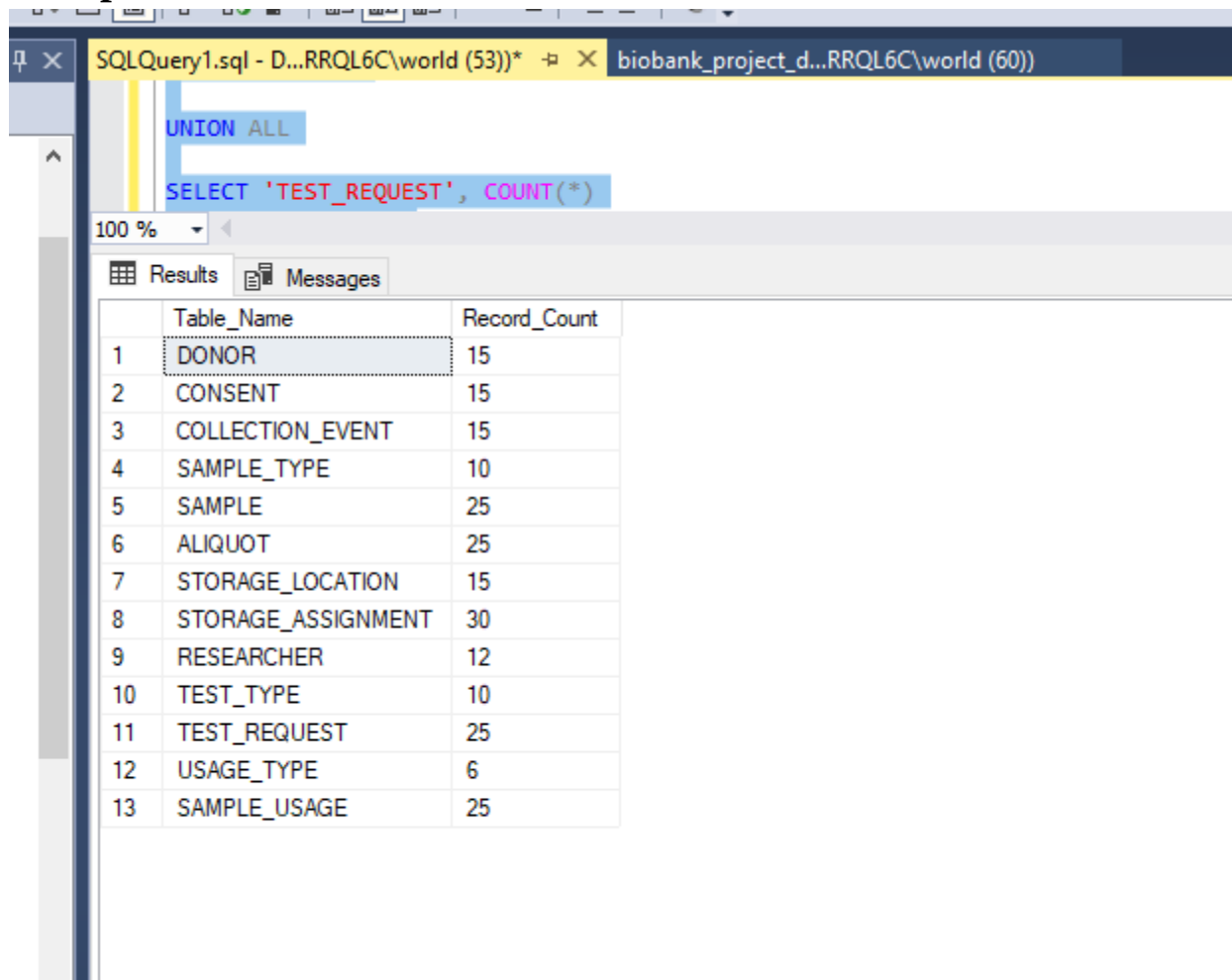
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Third Normal Form (3NF): Every non-key attribute in each relation describes only the entity identified by that relation's primary key, and not another non-key attribute. Sample-type properties (Type_Name, Description, Default_Storage_Temperature) belong to SAMPLE_TYPE and not to SAMPLE; test-definition properties belong to TEST_TYPE and not to TEST_REQUEST; storage properties (Facility, Freezer, Rack, Box, Position, Temperature) belong to STORAGE_LOCATION and not to STORAGE_ASSIGNMENT. No transitive dependency exists between a non-key attribute and another non-key attribute in the same relation, so every table in the schema satisfies 3NF. This separation reduces redundancy and eliminates insertion, update, and deletion anomalies.

The 3NF explanation is based on the functional dependencies implied by the entity structure and the primary/foreign-key and constraint design described above: within every relation, all non-key attributes depend on the whole primary key, on nothing but the primary key, and not on each other. No further decomposition is required for any of the 13 relations.

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8. Sample Data and Execution Evidence



The screenshot shows a SQL query window with the following text:

```

UNION ALL
SELECT 'TEST_REQUEST', COUNT(*)

```

The Results window displays the following data:

	Table_Name	Record_Count
1	DONOR	15
2	CONSENT	15
3	COLLECTION_EVENT	15
4	SAMPLE_TYPE	10
5	SAMPLE	25
6	ALQUOT	25
7	STORAGE_LOCATION	15
8	STORAGE_ASSIGNMENT	30
9	RESEARCHER	12
10	TEST_TYPE	10
11	TEST_REQUEST	25
12	USAGE_TYPE	6
13	SAMPLE_USAGE	25

Figure 7. Record Count Verification for All Database Tables.

Figure 7 shows the expected initial record count for each table after the runs.

The query script temporarily inserts RES013, updates it, and then deletes it. Therefore, after that demonstration the RESEARCHER table returns to its original 12 records. The invalid trigger and constraint tests are also designed to be rejected and should not add permanent records.

8.1 SQL Queries and Reporting

With the sample data in (Figure 7), the retrieval, join, aggregation, and subquery operations presented in Section 6 execute successfully against real donor, sample, and researcher records rather than an empty schema. These queries, together with the vw_DonorSampleSummary and vw_TestRequestDetails views described in

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Section 9.1, form the reporting layer of the system: they let a user retrieve donor-level sample counts, researcher workloads, and test-request status without writing raw joins each time.

8.2 CRUD Demonstration

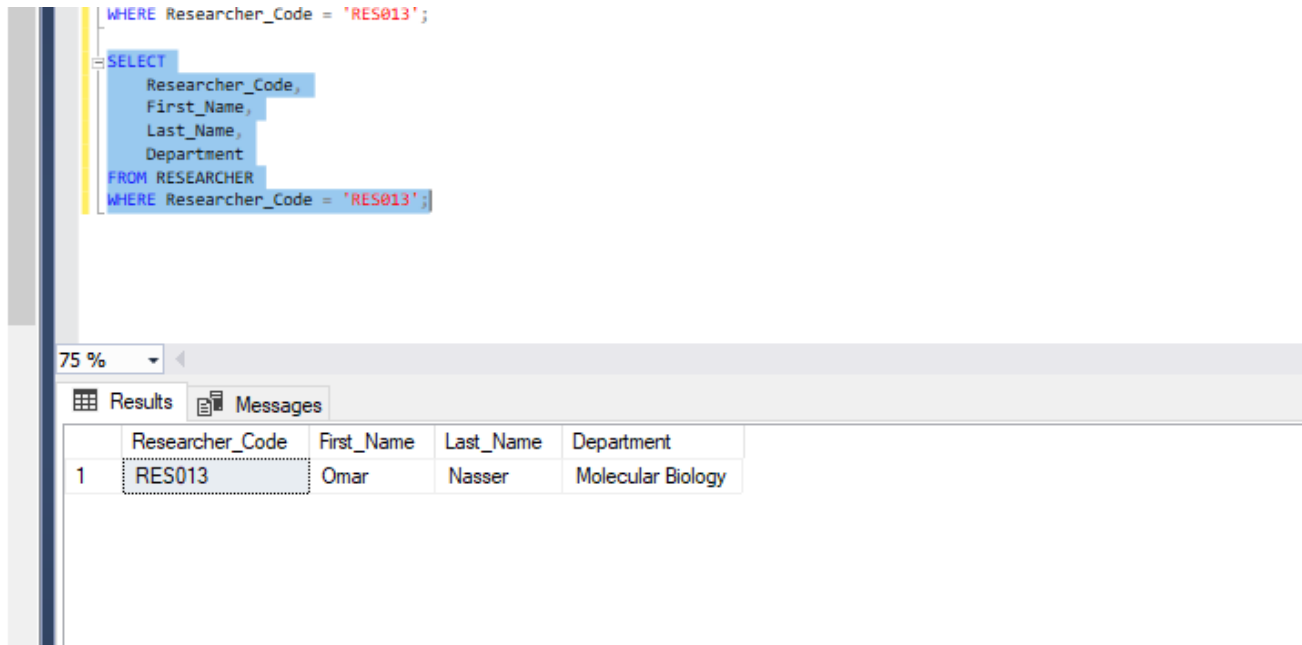


Figure 8. UPDATE Operation Result Showing the Modified Researcher Department.

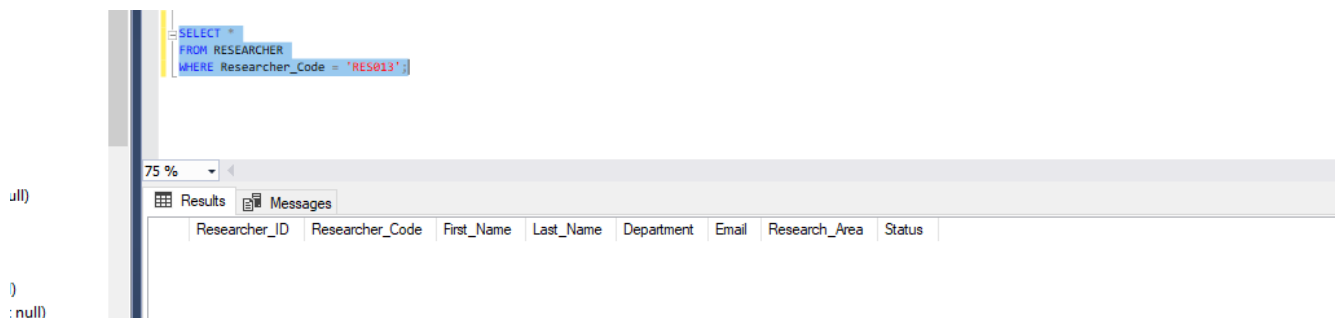


Figure 9. DELETE Operation Result Confirming the Removal of Researcher RES013.

Figures 8 and 9 demonstrate INSERT, UPDATE, DELETE, and SELECT operations using researcher RES013. The researcher is inserted, the department is changed from Genomics to Molecular Biology, and the researcher is then deleted. Verification SELECT statements confirm each stage.

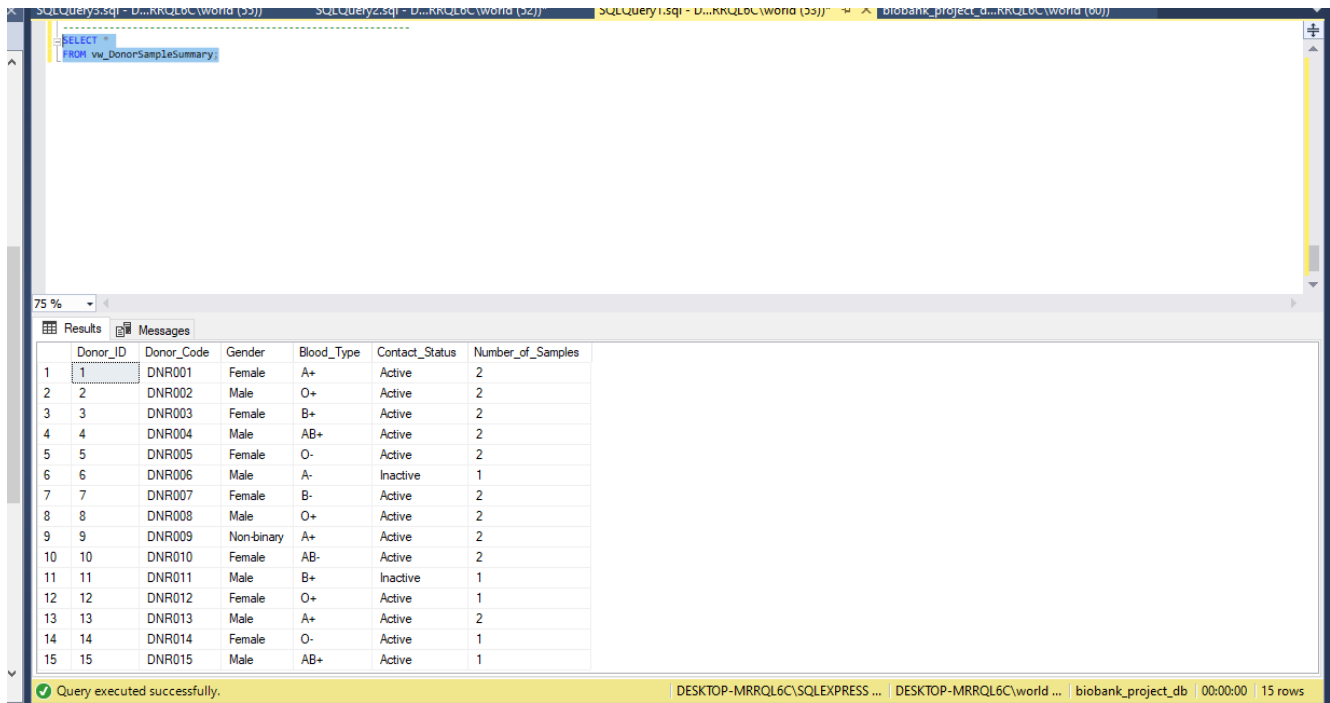
Operation	Demonstration
INSERT	Add RES013.

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UPDATE	Change RES013 department to Molecular Biology.
DELETE	Remove RES013.
SELECT	Verify the results.

9. Views, Indexes, and Business-Rule Trigger

9.1 Views



The screenshot shows a SQL query window with the following query:

```
SELECT *
FROM vw_DonorSampleSummary;
```

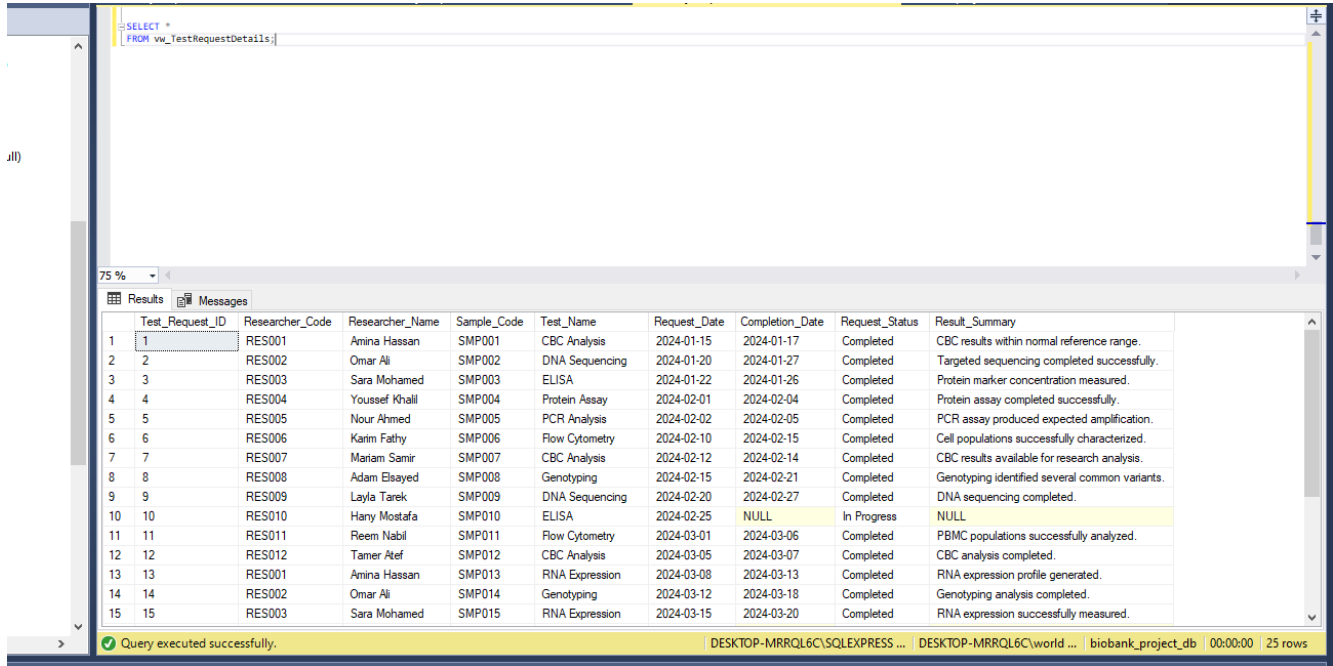
The results pane displays the following data:

Donor_ID	Donor_Code	Gender	Blood_Type	Contact_Status	Number_of_Samples
1	DNR001	Female	A+	Active	2
2	DNR002	Male	O+	Active	2
3	DNR003	Female	B+	Active	2
4	DNR004	Male	AB+	Active	2
5	DNR005	Female	O-	Active	2
6	DNR006	Male	A-	Inactive	1
7	DNR007	Female	B-	Active	2
8	DNR008	Male	O+	Active	2
9	DNR009	Non-binary	A+	Active	2
10	DNR010	Female	AB-	Active	2
11	DNR011	Male	B+	Inactive	1
12	DNR012	Female	O+	Active	1
13	DNR013	Male	A+	Active	2
14	DNR014	Female	O-	Active	1
15	DNR015	Male	AB+	Active	1

The status bar at the bottom indicates: Query executed successfully. | DESKTOP-MRRQL6C\SQLEXPRESS ... | DESKTOP-MRRQL6C\world ... | biobank_project_db | 00:00:00 | 15 rows

Figure 10. vw_DonorSampleSummary View Showing the Number of Samples Associated with Each Donor.

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Test_Request_ID	Researcher_Code	Researcher_Name	Sample_Code	Test_Name	Request_Date	Completion_Date	Request_Status	Result_Summary
1	RES001	Amina Hassan	SMP001	CBC Analysis	2024-01-15	2024-01-17	Completed	CBC results within normal reference range.
2	RES002	Omar Ali	SMP002	DNA Sequencing	2024-01-20	2024-01-27	Completed	Targeted sequencing completed successfully.
3	RES003	Sara Mohamed	SMP003	ELISA	2024-01-22	2024-01-26	Completed	Protein marker concentration measured.
4	RES004	Youssef Khaili	SMP004	Protein Assay	2024-02-01	2024-02-04	Completed	Protein assay completed successfully.
5	RES005	Nour Ahmed	SMP005	PCR Analysis	2024-02-02	2024-02-05	Completed	PCR assay produced expected amplification.
6	RES006	Karim Fathy	SMP006	Flow Cytometry	2024-02-10	2024-02-15	Completed	Cell populations successfully characterized.
7	RES007	Mariam Samir	SMP007	CBC Analysis	2024-02-12	2024-02-14	Completed	CBC results available for research analysis.
8	RES008	Adam Elsayed	SMP008	Genotyping	2024-02-15	2024-02-21	Completed	Genotyping identified several common variants.
9	RES009	Layla Tarek	SMP009	DNA Sequencing	2024-02-20	2024-02-27	Completed	DNA sequencing completed.
10	RES010	Hany Mostafa	SMP010	ELISA	2024-02-25	NULL	In Progress	NULL
11	RES011	Reem Nabil	SMP011	Flow Cytometry	2024-03-01	2024-03-06	Completed	PBMC populations successfully analyzed.
12	RES012	Tamer Atef	SMP012	CBC Analysis	2024-03-05	2024-03-07	Completed	CBC analysis completed.
13	RES001	Amina Hassan	SMP013	RNA Expression	2024-03-08	2024-03-13	Completed	RNA expression profile generated.
14	RES002	Omar Ali	SMP014	Genotyping	2024-03-12	2024-03-18	Completed	Genotyping analysis completed.
15	RES003	Sara Mohamed	SMP015	RNA Expression	2024-03-15	2024-03-20	Completed	RNA expression successfully measured.

Figure 11. vw_TestRequestDetails View Showing Detailed Test Request Information.

vw_DonorSampleSummary provides donor-level sample counts. vw_TestRequestDetails combines test requests with researcher, sample, and test type information for reporting.

9.2 Indexes

- IX_COLLECTION_EVENT_Donor_ID improves collection-event lookup by donor.
- IX_SAMPLE_Sample_Type_ID improves sample lookup by sample type.
- IX_TEST_REQUEST_Researcher_ID improves test-request lookup by researcher.

9.3 Business-Rule Trigger

trg_CheckSampleUsageQuantity fires after INSERT or UPDATE on SAMPLE_USAGE. It sums usage for affected samples and compares the total with the original SAMPLE.Collection_Volume. If the total exceeds the collection volume, the trigger raises an error and rolls back the transaction.

This rule is implemented as a trigger because it compares a row with the aggregate of multiple usage records, which cannot be enforced by a simple row-level CHECK constraint.

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10. Test Cases and Results

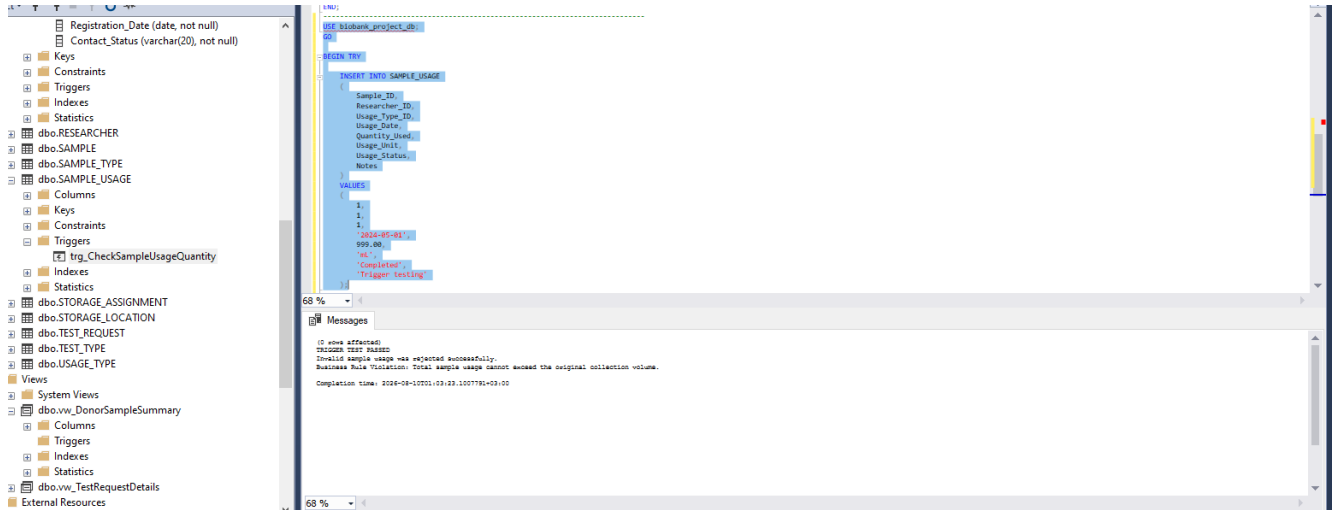


Figure 12. Trigger Test Result Confirming That Excessive Sample Usage Was Rejected.

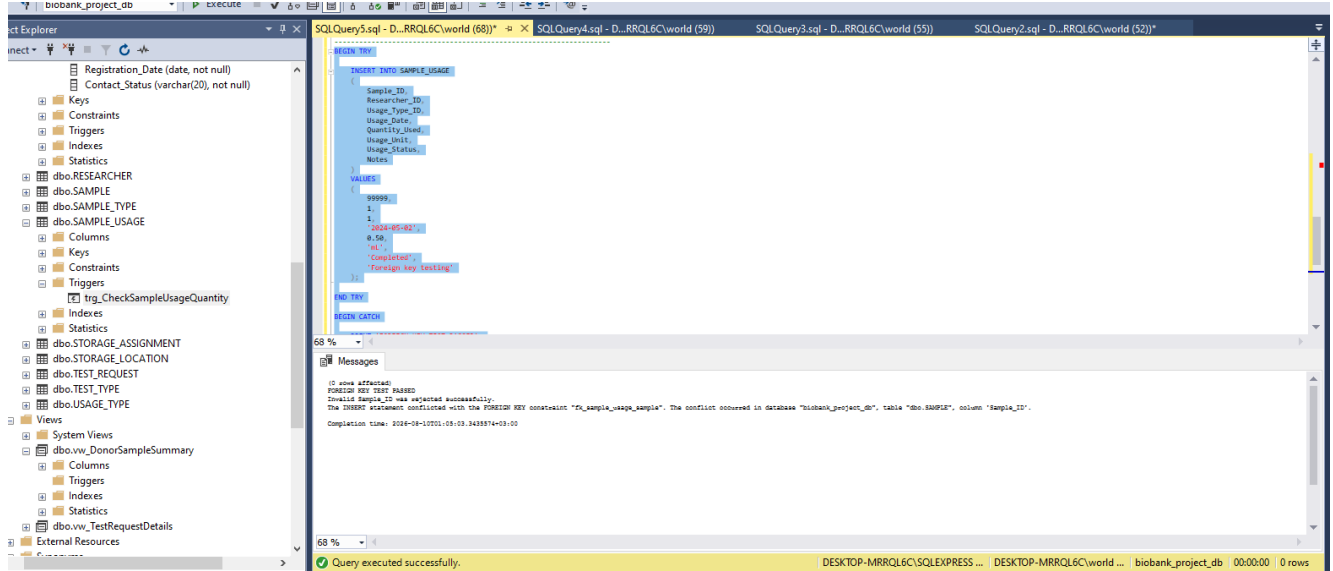
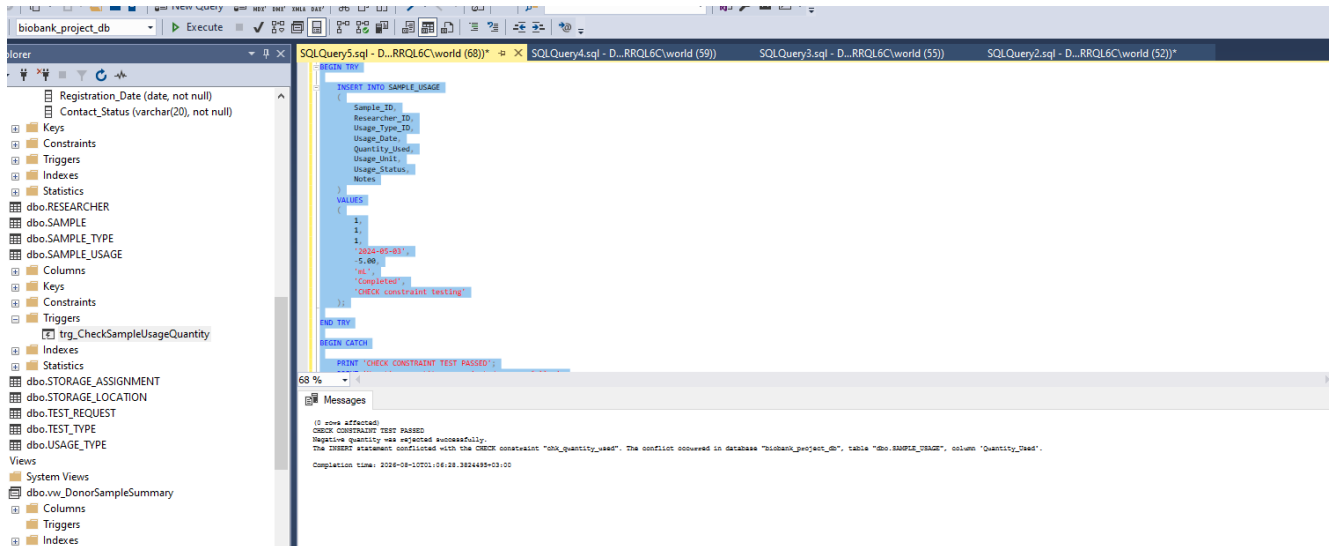


Figure 13. Foreign Key Constraint Test Result Confirming That an Invalid Sample ID Was Rejected.

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The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the database structure for 'biobank_project_db', including tables like 'dbo.RESEARCHER', 'dbo.SAMPLE', 'dbo.SAMPLE_TYPE', 'dbo.SAMPLE_USAGE', and 'dbo.STORAGE_ASSIGNMENT'. The right pane shows the execution of a SQL query (SQLQuery5.sql) that tests a CHECK constraint. The query includes an INSERT statement for 'SAMPLE_USAGE' and a PRINT statement to confirm the test result. The Messages pane at the bottom shows the output: 'CHECK CONSTRAINT TEST PASSED' and a message indicating that a negative quantity was rejected successfully.

```

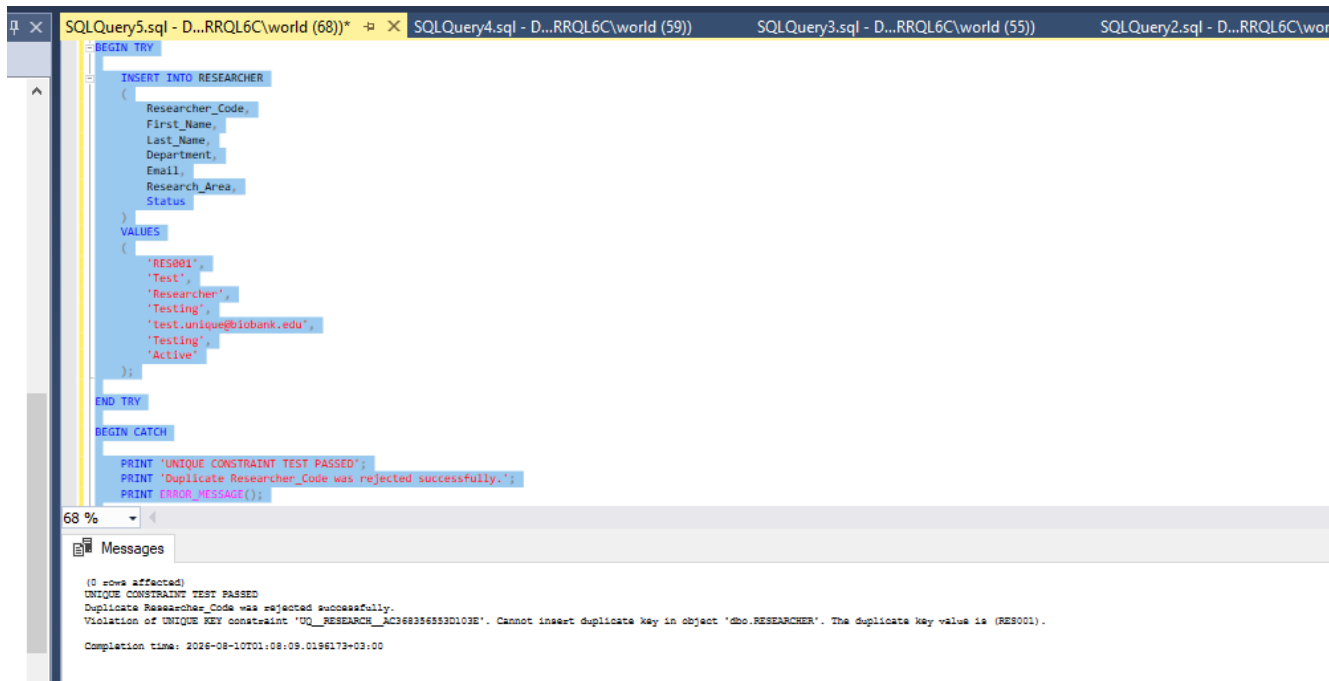
--BEGIN TRY
--INSERT INTO SAMPLE_USAGE
--(
--    Sample_ID,
--    Researcher_ID,
--    Usage_Type_ID,
--    Usage_Date,
--    Quantity_Used,
--    Usage_Unit,
--    Usage_Status,
--    Notes
--)
--VALUES
--(
--    1,
--    1,
--    1,
--    '2024-05-01',
--    -5.00,
--    'ml',
--    'Completed',
--    'CHECK constraint testing'
--);
--END TRY
--BEGIN CATCH
--PRINT 'CHECK CONSTRAINT TEST PASSED';
--END CATCH

```

Messages

(0 rows affected)
CHECK CONSTRAINT TEST PASSED
Negative quantity was rejected successfully.
The INSERT statement conflicted with the CHECK constraint "chk_quantity_used". The conflict occurred in database "biobank_project_db", table "dbo.SAMPLE_USAGE", column "Quantity_Used".
Completion time: 2024-08-10T01:06:26.3824489+03:00

Figure 14. CHECK Constraint Test Result Confirming That Negative Sample Usage Quantity Was Rejected.



The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the database structure for 'biobank_project_db'. The right pane shows the execution of a SQL query (SQLQuery5.sql) that tests a UNIQUE constraint. The query includes an INSERT statement for 'RESEARCHER' and PRINT statements to confirm the test result. The Messages pane at the bottom shows the output: 'UNIQUE CONSTRAINT TEST PASSED' and a message indicating that a duplicate Researcher Code was rejected successfully.

```

--BEGIN TRY
--INSERT INTO RESEARCHER
--(
--    Researcher_Code,
--    First_Name,
--    Last_Name,
--    Department,
--    Email,
--    Research_Area,
--    Status
--)
--VALUES
--(
--    'RES001',
--    'Test',
--    'Researcher',
--    'Testing',
--    'test.unique@biobank.edu',
--    'Testing',
--    'Active'
--);
--END TRY
--BEGIN CATCH
--PRINT 'UNIQUE CONSTRAINT TEST PASSED';
--PRINT 'Duplicate Researcher_Code was rejected successfully.';
--PRINT ERROR_MESSAGE();
--END CATCH

```

Messages

(0 rows affected)
UNIQUE CONSTRAINT TEST PASSED
Duplicate Researcher_Code was rejected successfully.
Violation of UNIQUE KEY constraint 'UQ_RESEARCHER_ACD369356553D103E'. Cannot insert duplicate key in object 'dbo.RESEARCHER'. The duplicate key value is (RES001).
Completion time: 2024-08-10T01:08:09.0196173+03:00

Figure 15. UNIQUE Constraint Test Result Confirming That a Duplicate Researcher Code Was Rejected.

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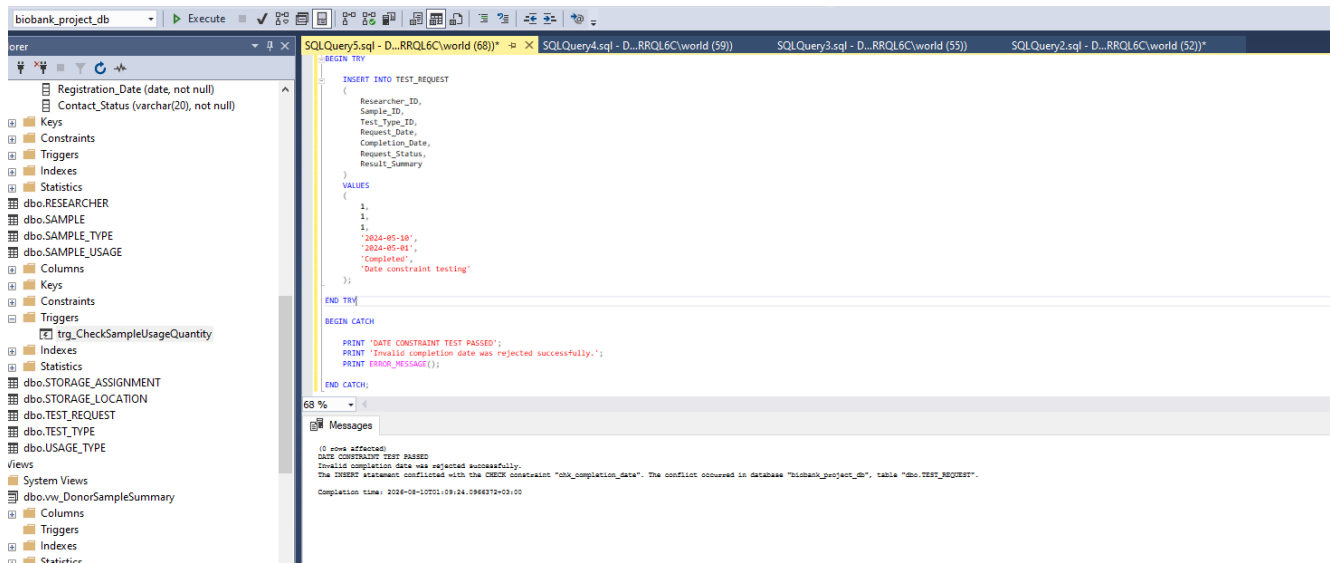


Figure 16. Date CHECK Constraint Test Result Confirming That an Invalid Completion Date Was Rejected.

The table below summarizes five negative test cases that cover the trigger and the principal integrity constraints.

Test	Invalid action	Expected result
Trigger	Insert 999.00 mL for Sample 1.	Rejected; transaction rolled back.
Foreign key	Use Sample_ID 99999.	Rejected because the sample does not exist.
CHECK	Insert Quantity_Used = -5.00.	Rejected because quantity must be greater than zero.
UNIQUE	Insert duplicate Researcher_Code RES001.	Rejected because the code is unique.
Date CHECK	Set Completion_Date before Request_Date.	Rejected by the date constraint.

The TRY/CATCH blocks print confirmation messages such as TRIGGER TEST PASSED, FOREIGN KEY TEST PASSED, CHECK CONSTRAINT TEST PASSED, UNIQUE CONSTRAINT TEST PASSED, and DATE CONSTRAINT TEST PASSED when the invalid operation is correctly rejected.

11. System Strengths, Limitations, and Future Improvements

- Clear separation of donor, consent, collection, specimen, storage, research, testing, and usage responsibilities.
- Strong referential integrity through foreign keys.

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- Domain validation through CHECK constraints.
- A trigger enforces the sample-volume business rule.
- Storage assignment history is supported.
- Views simplify recurring reports.
- Indexes support common foreign-key searches.

11.1 Recommended Future Improvements

- Add a filtered UNIQUE index so only one current storage assignment can exist for each aliquot.
- Add stronger consistency rules between Consent_Status and Withdrawal_Date.
- Validate measurement units before comparing or aggregating quantities.
- Consider linking SAMPLE_USAGE to ALIQUOT if exact aliquot-level consumption must be tracked.
- Add audit fields and role-based security for production or regulated use.
- Apply appropriate privacy controls for donor-related information.

12. Conclusion

The Biobank & Biospecimen Management System provides a complete relational foundation for managing biospecimens from donor registration and consent through collection, sample creation, aliquoting, storage, laboratory testing, and research usage.

The SQL demonstrates the required database construction, data population, querying, CRUD operations, indexing, views, business-rule enforcement, and integrity testing. Together with the ER diagram, the implementation provides a coherent database design suitable for further development.

13. References

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